Lab 2 Report

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1 Part 1: Spectrogram of a Chirp

In this part, I analyzed a linear chirp (100 Hz to 7000 Hz). I used three windows: Boxcar, Hann, and Hamming.

The Boxcar window produced the most "fuzzy" line with vertical smearing. This is because the abrupt edges of the rectangular window create leakage. The Hann window was the smoothest and cleanest because it tapers the signal perfectly to zero at the edges. The Hamming window was mostly clean but still showed vertical "cuts".

Window shape determines the trade-off between the sharpness of the line and noise. Boxcar is mathematically the sharpest but the messiest. Hann is cleaner but looks slightly "fatter." Window length determines the time-frequency trade-off: a long window gives great frequency resolution but smears the timing. A short window gives exact timing but blurry frequency results.

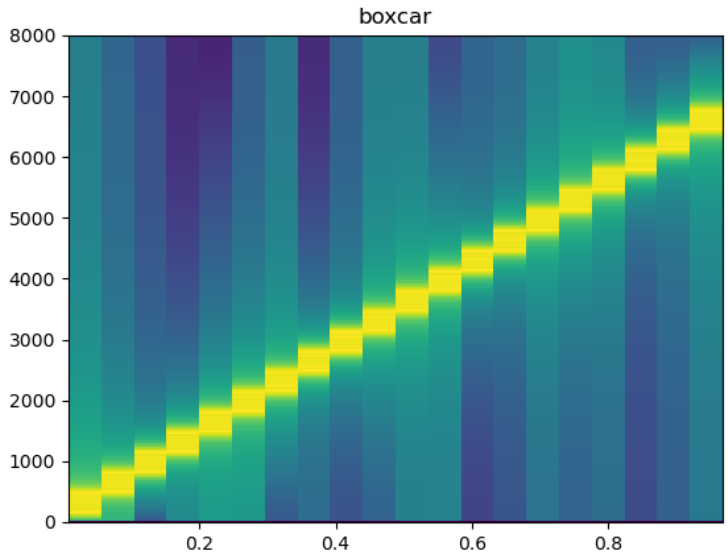

Figure 1: Spectrogram using a Boxcar Window.

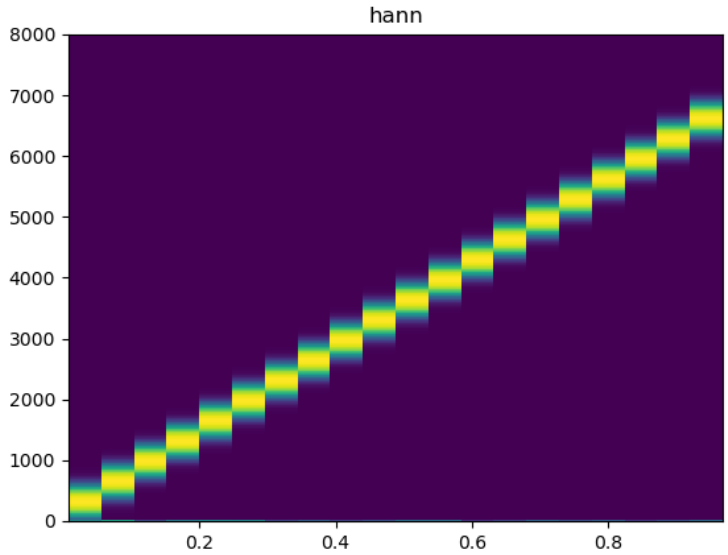

Figure 2: Spectrogram using a Hann Window.

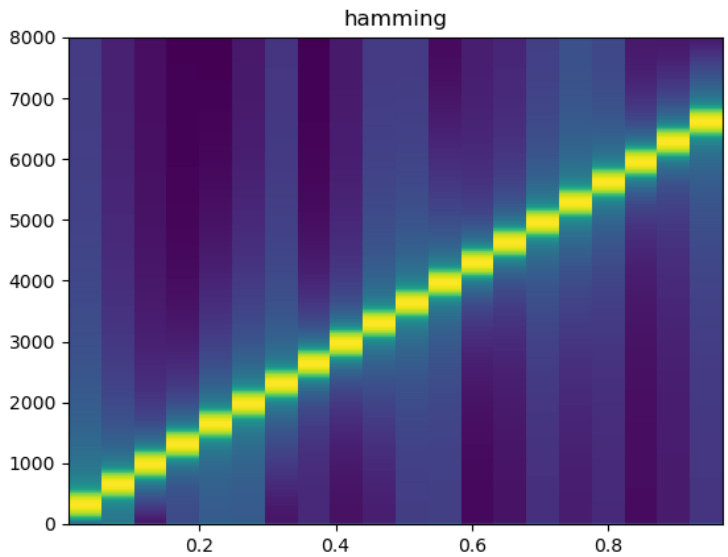

Figure 3: Spectrogram using a Hamming Window.

2 Part 2: Real Audio Spectrogram and Spectral Gain

I analyzed Beethoven's Tempest Sonata. I modified the magnitudes using Gamma factors and reconstructed the audio using the Inverse STFT.

The error was near 10^{-15}, proving the process is perfectly reversible.

I used a Gamma factor (γ) to adjust the loudness of different frequency components in the audio. Because the magnitudes are represented as fractions between 0 and 1, setting $\gamma = 0.8$ makes the quieter parts of the signal louder, which compresses the overall dynamic range. Conversely, applying $\gamma = 1.2$ results in dynamic range expansion, which attenuates these low-energy components. In the resulting visualizations, the $\gamma = 1.2$ waveform appears as a constant or flat line because the spectral expansion has suppressed the low-level signal information below the detectable visual threshold relative to the compressed $\gamma = 0.8$ signal.

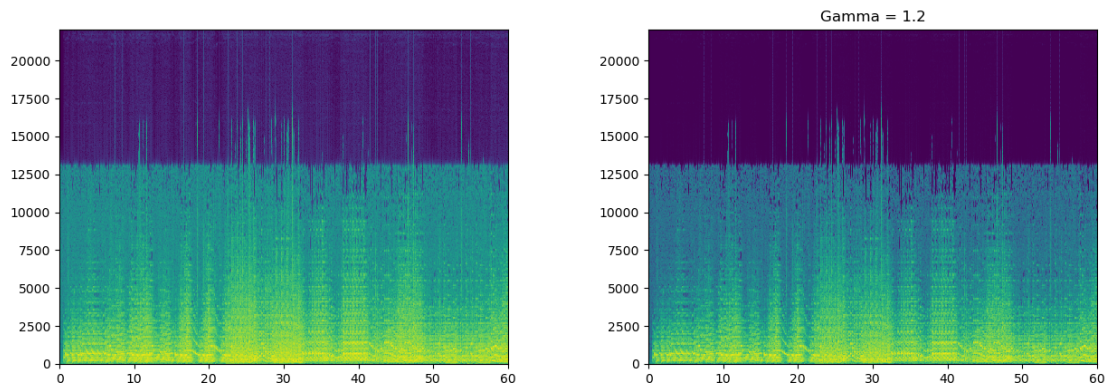

Figure 4: Comparison: Original Spectrogram (left) and Modified Gamma = 1.2 Spectrogram (right).

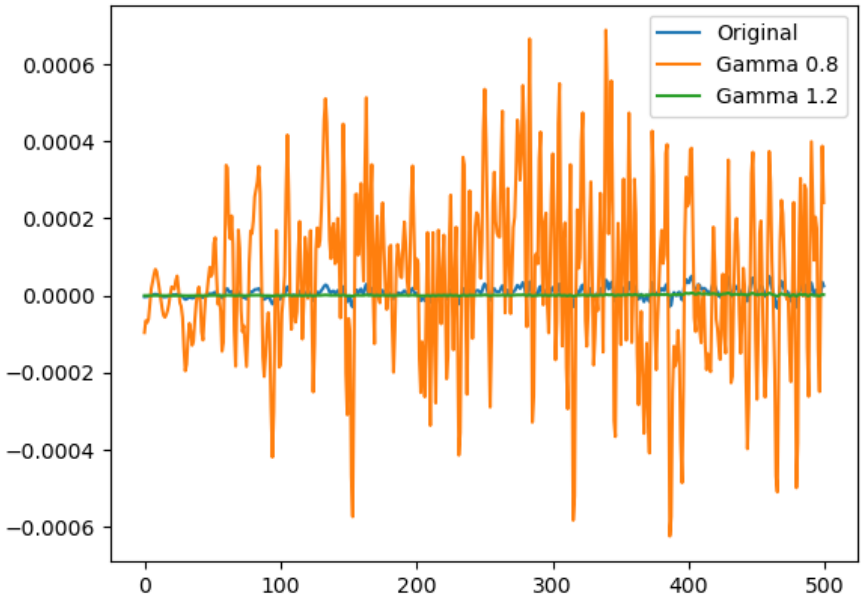

Figure 5: Time-domain plot comparing Original, Gamma 0.8, and Gamma 1.2 signals.

3 Part 3: AM vs FM Modulation Spectra

The AM spectrum has only three peaks: the carrier and two sidebands. The FM spectrum is much wider with many sidebands. This demonstrates that FM occupies more bandwidth than AM for the same message signal.

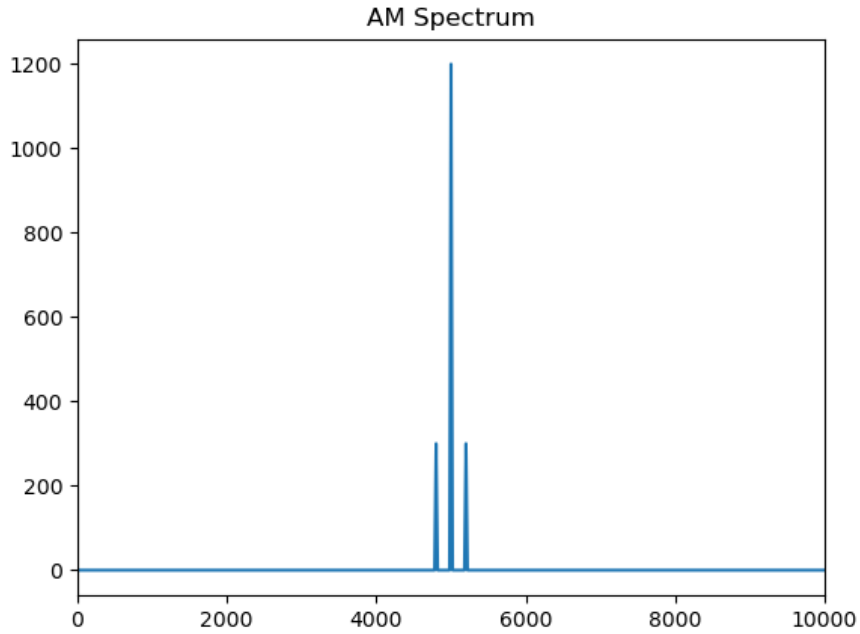

Figure 6: Magnitude Spectrum of Amplitude Modulation (AM).

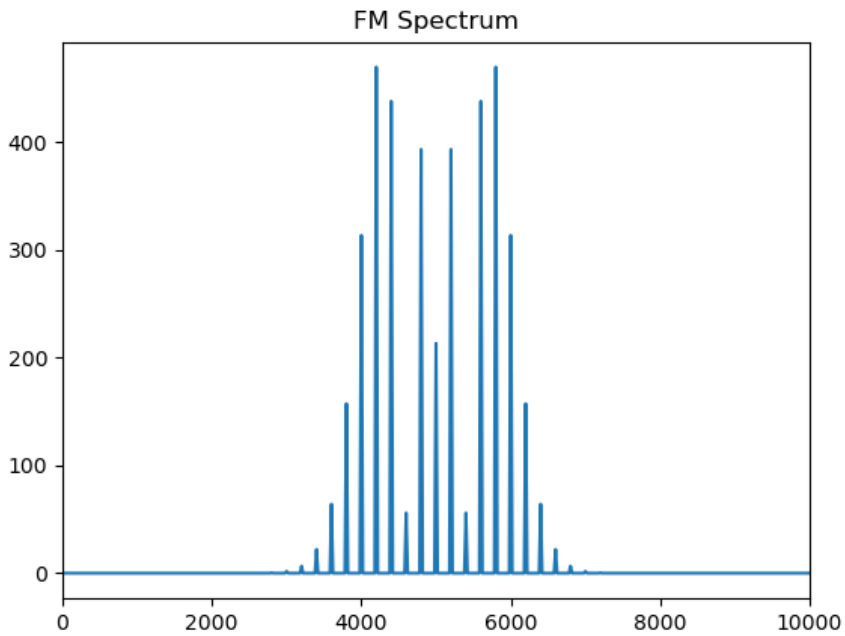

Figure 7: Magnitude Spectrum of Frequency Modulation (FM).

4 Part 4: Voiced vs Unvoiced Features

Using a recording of "zzzz" followed by "ssss", I calculated Energy and Zero-Crossing Rate (ZCR).

The "zzzz" region is voiced; it shows high Energy from vocal cord vibration and low ZCR because the wave is a low-frequency periodic signal. The "ssss" region is unvoiced; it shows low Energy and high ZCR because it is high-frequency noise created by air turbulence at the teeth.

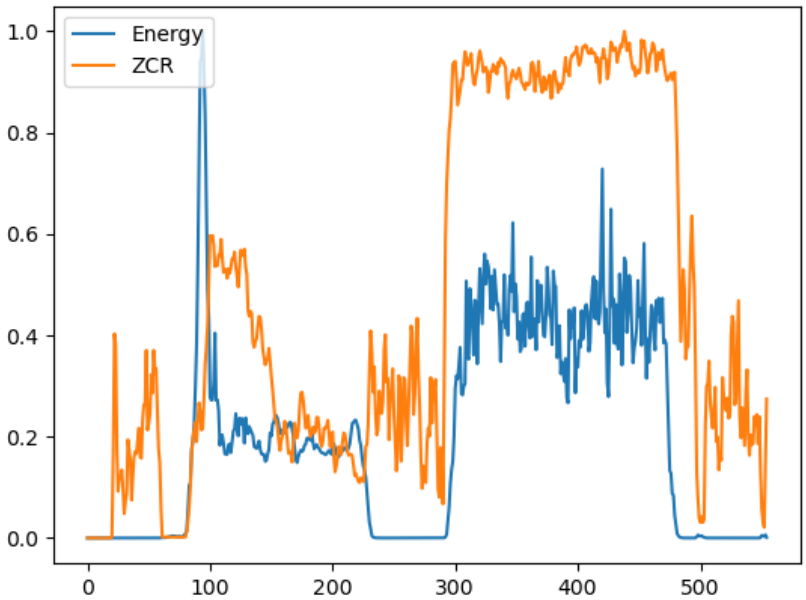

Figure 8: Normalized Energy and Zero-Crossing Rate (ZCR) for the voiced and unvoiced segments.

5 Part 5: Fourier Uncertainty

I used Gaussian packets to test the uncertainty principle $\Delta x \Delta k \geq 0.5$.

The analysis of Gaussian wave packets reveals a clear inverse relationship: As σ_x decreases and the pulse becomes more localized in position space, the width in k-space (Δk) increases proportionally. For every Gaussian case tested, the product $\Delta x \Delta k$ remained approximately 0.5000.

By using the physical relation $p = \hbar k$ with $\hbar = 1$, I verified that the momentum uncertainty Δp is numerically equivalent to Δk in this system. In every simulation, the product $\Delta x \Delta p$ was found to be greater than or equal to 0.5, which directly satisfies the inequality $\Delta x \Delta p \geq \hbar/2$. This experiment relates directly to Heisenberg's Uncertainty Principle. Because position and momentum are Fourier transform pairs, a signal cannot be infinitely localized in both domains simultaneously; sharpening the signal in one domain inevitably broadens it in the other.

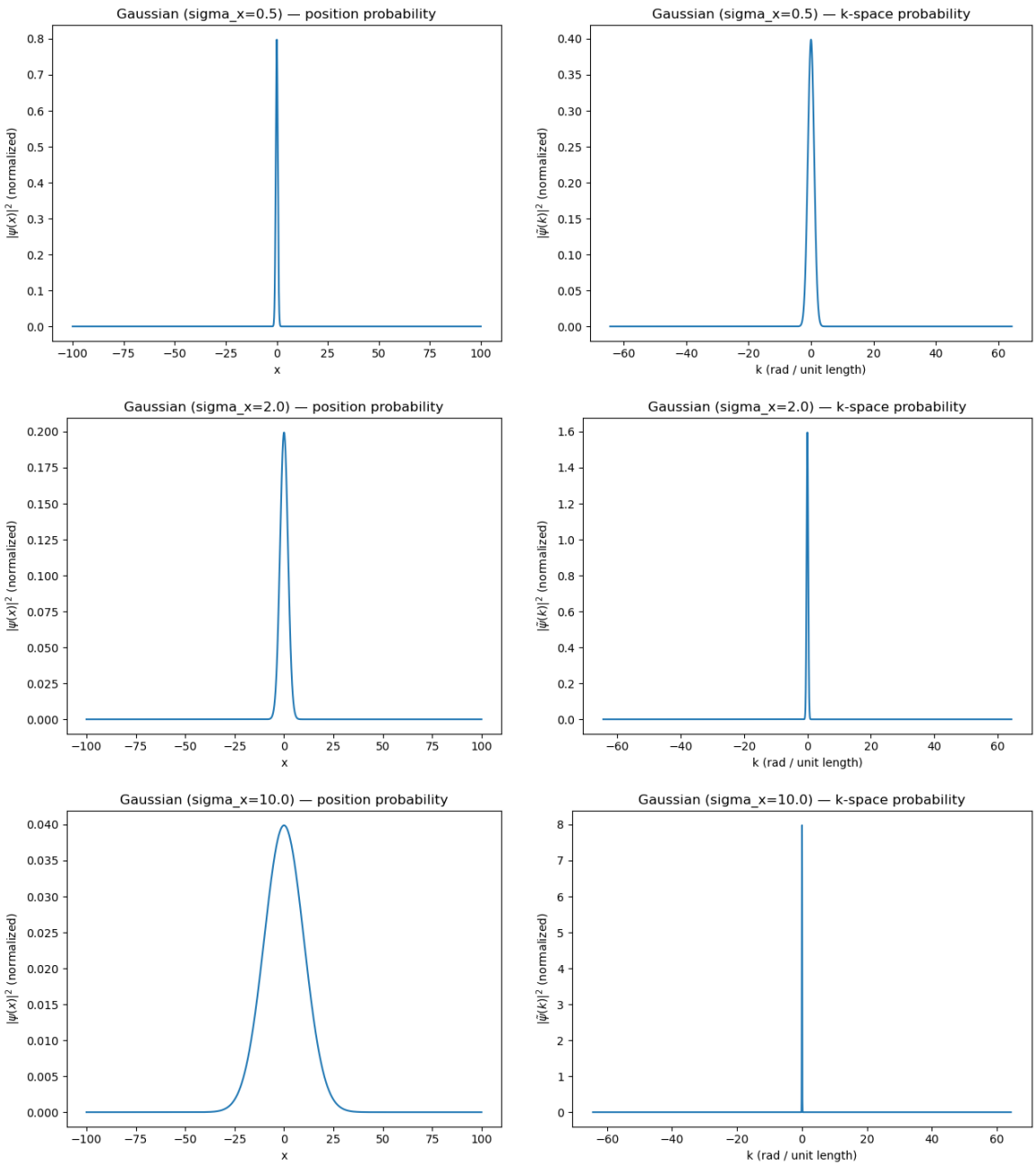

Figure 9: Position space (left) and k-space (right) probability densities for $\sigma_x = 0.5$.